Satyam Awasthi

Researcher & Software Engineer

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Professional Summary

Computer Science researcher specializing in the intersection of **Autonomous Decision-Making**, **Spatial Perception**, and **Robot Learning** within embodied systems. My research investigates how perception, planning, and learning can be integrated to enable robust and verifiably safe execution in complex autonomous agents.

Research Interests

Embodied AI, Autonomous Systems, and Robot Learning with emphasis on integrating perception, planning, and decision-making for safe and reliable long-horizon autonomy.

Education

University of California, Santa Barbara (UCSB) Santa Barbara, CA *M.S. in Computer Science (GPA: 3.91/4.00)* Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur) Kharagpur, India *B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0)* Jul 2016 – Jul 2020

Research Experience

Four Eyes Laboratory, UC Santa Barbara Sep 2021 – Jun 2024 *Graduate Researcher* Advisor: Prof. Tobias H"ollerer
Co-Advisor: Prof. Michael Beyeler

- Developed EyeTTS, a calibration and evaluation framework for eye tracking in mixed-reality locomotion.
- Designed experimental protocols and software pipelines to quantify gaze-tracking drift during active user motion.
- Curated a multi-device eye-tracking dataset and conducted user studies across diverse locomotion layouts.
- Published first-author poster papers at IEEE VR and IEEE ISMAR Adjunct venues.

Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur Jan 2016 – Dec 2018 *Undergraduate Researcher*

- Developed a hardware-agnostic navigation stack integrating EKF-based localization from LiDAR, IMU, and GPS streams.
- Implemented visual obstacle segmentation using a custom U-Net architecture and hybrid motion planning using A*/Dijkstra and Dynamic Window Approach.
- Designed decentralized multi-agent coordination using relative-pose estimation over an ad-hoc mesh network.
- Enabled zero-shot Sim2Real transfer across heterogeneous robotic platforms.

Publications

Peer-Reviewed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing *Proceedings of the 14th International Conference on COMmunication Systems & NETworkS (COMSNETS)*, 2022.
DOI: 10.1109/COMSNETS53615.2022.9668531
Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra. *COMSNETS*, 2022.

Workshop & Poster Publications

Eye Tracking Performance in Mobile Mixed Reality *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 2024.
DOI: 10.1109/VRW62533.2024.00321
Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias H"ollerer. *IEEE VRW*, 2024.

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, 2023.
DOI: 10.1109/ISMAR-Adjunct60411.2023.00104
Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias H"ollerer. *ISMAR Adjunct*, 2023.

Research Artifacts

EyeTTS Calibration Framework 2022–2024 Calibration Framework · User Study Framework

- Developed an extensible framework for evaluating and calibrating eye-tracking systems during mixed-reality locomotion.
- Implemented device-agnostic benchmarking pipelines and automated drift-correction procedures.
- Released reproducible software infrastructure supporting eye-tracking research.

EyeTTS Mixed-Reality Eye Tracking Dataset 2022–2024 Dataset · User Study Videos

- Curated a multi-device mixed-reality locomotion dataset for calibration and performance evaluation.
- Collected gaze trajectories, calibration measurements, and locomotion metadata across diverse MR layouts.
- Supported experimental validation reported in ISMAR and VRW publications.

Industry Experience

Intuit Inc Mountain View, CA · Mar 2024 – Jan 2026
Software Engineer 2

- Built dual-stage **agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc Mountain View, CA · Apr 2023 – Nov 2023
Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.
- Designed a dynamic, state-driven carousel UI framework optimized for client-side rendering efficiency.

Coupang Global LLC Mountain View, CA · Jun 2022 – Sep 2022
Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute Noida, India · Sep 2020 – Sep 2021
Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Selected Projects

JitterNot Jan 2022 – Mar 2022
LSTM-Based Model Predictive Controller for Adaptive Video Streaming UCSB

- Developed an LSTM network for real-time throughput prediction, transforming the closed-loop control system from MISO to SISO.
- Implemented an MPC controller leveraging Receding Horizon Control (RHC) principles to dynamically calculate optimal video resolution.

In Motion — HAR using DeepConvLSTM Nov 2019 – Nov 2019
Distributed Mobile Computing System for Human Activity Recognition IIT Kharagpur

- Designed a client-server telemetry pipeline streaming high-frequency IMU sensor arrays for rolling window inference.
- Implemented a DeepConvLSTM core architecture using TimeDistributed CNN layers to model spatial-temporal features efficiently.

CohesiveAR Apr 2022 – Jun 2022
Augmented Reality Pipeline for Texture Extraction and Homography Mapping UCSB

- Developed a spatial perception pipeline using Google ARCore and OpenCV to map and project dynamic 3D scene geometry into unified 2D reference frames.
- Implemented matrix homography and perspective rectification algorithms to correct for camera tilt skew, extracting invariant surface textures from variable viewpoints.

Tweeter: Distributed Infrastructure Testbed Sep 2021 – Dec 2021
High-Concurrency Microblogging Testbed on AWS EC2 Clusters UCSB

- Conducted high-throughput load testing up to 512 users/sec, mapping user trajectories as state machines to latency scaling curves.
- Optimized read throughput by deploying nested fragment caches and database query preloading to eliminate bottleneck points.

Teaching

University of California, Santa Barbara Santa Barbara, CA
Graduate Teaching Assistant Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Technical Skills

Programming: Python, C++, Java, Kotlin, MATLAB

Robotics & Simulation: ROS/ROS2, Gazebo, RViz, Unity

Machine Learning & Vision: PyTorch, OpenCV

Methods: State Estimation (EKF), Model Predictive Control (MPC), Motion Planning, Sensor Fusion, Spatial Perception

Developer Tools: Git

Academic Service & Peer Review

Technical Program Committee & Peer Reviewer
ACM Symposium on User Interface Software and Technology (UIST) May 2024
Served as an external peer reviewer evaluating full paper submissions for novel interface hardware, interactive system architectures, and spatial tracking methodologies.